Evaluating and Reducing Hallucinations in LLM-Based Medical Report Summarization

Second Progress Report, Fall 2026 (COMP 698C), revised after the October review

Srilaya Ponangi, M.S. Computer Science

Committee: Dr. Taehyung (George) Wang (chair), Dr. Robert McIlhenny, Dr. Mahdi Ebrahimi

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Goal and research questions (from the proposal)

- Goal: measure hallucinations in LLM-generated patient-facing summaries at the claim level with a judge validated against medical experts, and test whether retrieval grounding and verification reduce them on real clinical notes
- RQ1: how often are summary statements unsupported by, or contradicting, the source note?
- RQ2: which statements are unsupported, and how well does the automatic judge agree with medical experts?
- RQ3: do retrieval-augmented generation and verification reduce unsupported and contradicted statements, compared with an extractive reference, and at what cost in coverage?
- Three studies: a pilot on public MTSamples notes (GPT-4o-mini), a judge selection study against expert labels, and the main study on 110 MIMIC-IV hospital courses with an open-weight model running locally; the proposal's second task, question answering, is piloted with abstention

Proposal commitments and where they stand

Commitment in the proposal	Status
Claim-level NLI judge calibrated against expert labels	Done: pilot judge at chance against the experts (kappa 0.06); seven candidates compared; DeBERTa-v3-large NLI selected (kappa 0.33, AUROC 0.78); audited blind on the main-study outputs (kappa 0.20); checked on 12,614 clinician-labeled VeriFact propositions (kappa 0.25)
Experiments on MIMIC data	Done: 110 MIMIC-IV hospital courses; Qwen2.5-7B-Instruct served on the laptop; replicated with Llama-3.1-8B and Mistral-7B; no protected text leaves the machine
RAG, verification and an extractive baseline	Done in both studies: E0 full note, E1 RAG (excerpts), E1b RAG (course + excerpts), E3 RAG + Chain-of-Verification, E2 extractive
Coverage against a reference	Done: share of the clinician's discharge-instruction sentences that the summary supports (judge run in reverse)
Citation accuracy	Done: retrieval conditions cite an excerpt after every sentence; the judge checks each citation
Retrieval ablations (chunk size, top-k, BM25 vs dense, citations)	Done: six E1 variants on the same courses; no setting changed faithfulness after correction, coverage follows the amount of retrieved text
Question answering with abstention	Piloted (exploratory): three questions per course, full course vs retrieved excerpts, 'Not stated in the note.' allowed

Pilot on 50 public notes: the pipeline works, the small MiniLM judge does not

Group	Sentences	Expert-flagged	Judge-flagged	Precision	Recall	Kappa	AUROC
all	1,781	20.4%	84.1%	0.23	0.94	0.06	0.65
generated	805	12.5%	83.7%	0.14	0.96	0.04	0.68
doctor written	976	26.8%	84.3%	0.30	0.94	0.07	0.66
llama 70b original	159	26.4%	83.0%	0.30	0.95	0.10	0.67
gpt4 zero shot	151	6.0%	87.4%	0.06	0.89	0.00	0.50

Five conditions were compared on 50 MTSamples notes with GPT-4o-mini (UFR 0.81 for full-note summaries, 0.69 with verification, 0.12 for verbatim extracts), but on 1,781 expert-annotated sentences this judge flags 84% where the experts flag 20%: kappa 0.06 is chance level, and no evidence-aggregation variant raised it above 0.14. The pilot therefore establishes the pipeline and the artifacts it must handle (header lines, abstentions, citation markers), not hallucination rates.

Judge selection: seven candidates scored against the expert annotations (1,781 sentences)

Judge	Evidence	AUROC	Precision	Recall	Kappa	Flag rate
DeBERTa-v3-large NLI	whole course, windowed	0.78	0.44	0.54	0.33	25%
MiniCheck DeBERTa-v3-large	whole course, windowed	0.73	0.41	0.57	0.31	28%
DeBERTa-v3-large NLI fine-tuned on MedNLI	whole course, windowed	0.75	0.44	0.46	0.31	21%
Bespoke-MiniCheck-7B	top-3 sentences	0.75	0.39	0.60	0.30	31%
Bespoke-MiniCheck-7B	whole course, windowed	0.72	0.38	0.62	0.29	33%
DeBERTa-v3-large NLI	top-3 sentences	0.75	0.36	0.74	0.29	42%
DeBERTa-v3-large NLI fine-tuned on MedNLI	top-3 sentences	0.74	0.41	0.48	0.28	24%
MiniCheck DeBERTa-v3-large	top-3 sentences	0.73	0.34	0.68	0.25	41%
MiniCheck RoBERTa-large	whole course, windowed	0.72	0.41	0.36	0.24	18%
MiniCheck RoBERTa-large	top-3 sentences	0.70	0.33	0.60	0.22	37%
DeBERTa-v3-large cross-encoder NLI	top-3 sentences	0.69	0.28	0.85	0.17	62%
MiniLM cross-encoder	top-3 sentences	0.66	0.28	0.71	0.15	52%
DeBERTa-v3-large cross-encoder NLI	whole course, windowed	0.64	0.27	0.60	0.13	44%
MiniLM cross-encoder	whole course, windowed	0.58	0.24	0.68	0.07	58%

Rule fixed in advance: highest kappa on all sentences, threshold chosen on the other subset. Selected: DeBERTa-v3-large NLI (whole course, windowed), kappa 0.33 (0.30 under nested cross-validation), AUROC 0.78. Model size and training data matter more than the aggregation rule; MedNLI fine-tuning raised MedNLI accuracy but not agreement with the experts. Moderate agreement, so main-study rates are judge-estimated and the paired comparisons carry the weight.

External check: the selected judge on clinician-labeled propositions (VeriFact-BHC, Section 6.7)

Propositions	n	AUROC	Sensitivity	Specificity	Kappa
All propositions	12,614	0.82	0.29	0.92	0.25
Human-written summaries	8,175	0.79	0.30	0.89	0.22
LLM-written summaries	4,439	0.74	0.15	0.94	0.09
Claims	9,065	0.83	0.31	0.92	0.27
Sentences	3,549	0.77	0.27	0.89	0.19

Same judge, windowing and threshold as the main study, on the 5 best-matching 400-token windows of the admission's notes (63,070 judge calls; an admission spans 17 to 676 windows). Scoring every window instead, on the 13 shortest admissions, drops kappa from 0.37 to 0.00: the maximum over hundreds of windows supports almost everything. The best of the dataset's own 72 LLM rater configurations with the same reference reaches kappa 0.48. The judge still separates supported from unsupported statements on another institution's data, with the same low sensitivity the manual audit found.

Main study: data and design

- Corpus: 110 de-identified MIMIC-IV Brief Hospital Course sections (ann-pt-summ, PhysioNet), each paired with the discharge instructions the treating clinician wrote; the same dataset's 210 expert-annotated summaries validated the judge, and no patient is shared between the two
- Generator: Qwen2.5-7B-Instruct (4-bit) served on the laptop; the same prompts, decoding and retrieval rerun with Llama-3.1-8B-Instruct and Mistral-7B-Instruct-v0.3
- Conditions: E0 full note; E1 RAG with excerpts only; E1b RAG with the course and the excerpts; E3 RAG with Chain-of-Verification; E2 extractive; retrieval conditions cite the excerpt behind every sentence, and 'Not stated in the note.' is allowed
- Judge: DeBERTa-v3-large NLI, whole-course evidence in 400-token windows, $\tau = 0.20$; judge-estimated UFR and CR per summary; the clinicians' own instructions scored as a reference
- Measures: coverage of the clinician's instructions (judge in reverse), citation accuracy, six retrieval ablations, length-matched and no-citation variants, repeated generations, question-answering pilot
- Checks of the judge on these outputs: blind manual audit in two passes, second judge, second threshold, atomic facts, overlapping windows
- Ethics: MIMIC-IV-Note, ann-pt-summ, MedNLI and VeriFact-BHC are PhysioNet credentialed data (CITI training completed); every text is processed by models on the author's laptop, nothing is sent to an API or to cloud storage, and an automated scan finds no protected text in the repository

Main study results (n = 110 hospital courses; judge: DeBERTa-v3-large NLI, τ = 0.20)

Metric	E0 full-note	E1 RAG excerpts	E1b RAG + course	E3 RAG + CoVe	E2 extractive	Clinician-written
UFR mean (judge-estimated)	0.137	0.075	0.073	0.068	0.058	0.289
CR mean	0.031	0.030	0.032	0.031	0.053	0.080
Coverage of the clinician's sentences	54.3%	36.9%	53.2%	33.6%	22.5%	—
Claims cited / citation accuracy (pooled)	—	64% / 83%	21% / 50%	—	—	—
Abstentions per summary	0.0	0.0	0.0	0.7	0.0	0.0
Claims per summary	16.0	8.4	13.8	7.9	4.9	8.9
Words per summary	261	151	228	150	87	133

E1 vs E0: UFR $p < 0.001$, CR $p = 0.985$, coverage $p < 0.001$. E1b vs E0: UFR $p < 0.001$, CR $p = 0.462$, coverage $p = 0.449$. E3 vs E0: UFR $p < 0.001$, CR $p = 0.878$, coverage $p < 0.001$. E2 vs E0: UFR $p < 0.001$, CR $p = 0.169$, coverage $p < 0.001$. Paired two-sided Wilcoxon tests on the same courses; the primary family of twelve tests is Holm-adjusted in the thesis.

Main study: what the results say

- Grounding: E1, E1b and E3 lowered the judge-estimated UFR from 0.137 to 0.075, 0.073 and 0.068 (all $p < 0.001$); the three do not differ from one another; CR did not change
- Omission: E1 covered 37% of the clinician's sentences and E3 34%, against 54% for E0; E1b 53%, within 5 points of E0 by two one-sided tests
- References, not bounds: verbatim extracts (E2) score 0.058, judge error on supported text; the clinicians' own instructions 0.289, because they state what the course does not
- Brevity explains part of it, not all: an E0 told to write about 150 words scores 0.125 ($p = 0.405$ vs E0), still above the grounded conditions
- Citations: E1 cites 64% of its claims and 83% of those citations hold, E1b 21% and 50%; no retrieval setting changed faithfulness
- Question answering (exploratory): retrieval raised abstention from 23% to 33% and lowered the UFR of the answers given (0.429 to 0.307); warning-sign questions are rarely answerable from the course

Manual audit of the judge on the main-study claims (Section 7.9)

Condition	Labeled	Manual unsupported (95% CI)	Judge flags	Precision	False omission	Sensitivity	Specificity	Kappa
E0 full-note	83	23.1% (13.9% to 33.3%)	13.7%	0.52	0.18	0.32	0.91	0.27
E1 RAG excerpts	90	20.8% (9.9% to 32.0%)	7.3%	0.60	0.18	0.21	0.96	0.23
E1b RAG + course	90	8.2% (2.0% to 15.8%)	7.1%	0.29	0.07	0.25	0.94	0.21
E3 RAG + CoVe	70	21.7% (10.9% to 33.4%)	6.2%	0.46	0.20	0.13	0.96	0.11

The author labeled 333 claims blind to the condition and the judge's label, in two passes that agree at kappa 0.52 after the rubric was clarified. The judge understates unsupported content, mostly partly supported sentences. Manual differences from E0: E1 -2.2 points (-17.8 to +13.0); E1b -14.8 points (-26.8 to -2.8); E3 -1.3 points (-15.0 to +12.8). The interval excludes zero for E1b, which confirms the judge's direction there; the other differences are unresolved at this sample size.

Robustness: the same comparison under three generators (judge-estimated UFR, Section 7.10)

Condition	Qwen2.5-7B (main)	Llama-3.1-8B-Instruct	Mistral-7B-Instruct-v0.3
E0 full-note	0.137	0.161	0.133
E1 RAG excerpts	0.075	0.116	0.125
E1b RAG + course	0.073	0.101	0.110
E3 RAG + CoVe	0.068	0.105	0.164
Clinician-written	0.289	0.289	0.289

Under Llama-3.1-8B-Instruct every grounded condition is again below E0 after Holm adjustment; under Mistral-7B-Instruct-v0.3 no grounded condition differs from E0 after adjustment and E3 is higher. A second judge (Bespoke-MiniCheck-7B) agrees with the selected judge on 82% of claims (kappa 0.25) and keeps every grounded condition below E0; it puts E3 below E1 ($p < 0.001$) where the selected judge sees no difference, so whether verification adds to retrieval depends on the judge. The reductions also hold at threshold 0.5 and in each of three generations on 40 courses.

Hypotheses: verdicts across both studies

Hypothesis	Pilot (MTSamples, GPT-4o-mini)	Main study (MIMIC-IV, local models)	Verdict
H1: unsupported content is measurable, and the judge agrees with experts better than chance	UFR 0.811 under a judge with kappa 0.06 (chance level)	UFR 0.137 judge-estimated; 23.1% by manual reading; judge kappa 0.33	First half supported (confirmed by the audit); second half only moderately, for the selected judge
H2: RAG reduces unsupported and contradicted statements	CR down 24% ($p = 0.011$, Holm $p = 0.135$); UFR unchanged	UFR down 45 percent ($p < 0.001$); CR unchanged ($p = 0.985$)	Supported for unsupported facts on hospital courses with Qwen2.5-7B and Llama-3.1-8B, not with Mistral-7B; not for contradictions
H3: verification adds to RAG	E3 vs E1 UFR $p < 0.001$	Selected judge: E3 = E1 ($p = 0.212$); second judge: 0.079 vs 0.135 ($p < 0.001$)	Supported in the pilot; judge-dependent in the main study, so not claimed

Paired two-sided Wilcoxon tests on the same documents. Pilot effects describe a judge later shown invalid and are not interpreted further.

Thesis status and what changed since the October review

- Revised draft: 105 pages in CSUN format; 35 tables, 5 figures, 74 references; nine chapters (Introduction, Literature Review, Data, Methodology, Pilot Study, Selecting a Valid Judge, Main Study, Discussion, Conclusion) and Appendices A to C (prompts; code repository and reproduction; manual audit protocol); every table and figure is regenerated by one command from the result files
- Length: 163 to 105 pages by rewriting in the compact form of recent CSUN theses (each result a table or figure with a short paragraph; methods state what was done; the discussion interprets without repeating the results), with nothing removed from the evidence
- Judge-estimated versus confirmed rates: every rate is labeled judge-estimated; a blind manual audit in two passes measures the judge on these outputs and confirms the E1b reduction directly (Sections 4.10.5, 7.9)
- Judge evaluation: claim splitting corrected (citation markers, merged headers); selection optimism measured by nested cross-validation (kappa 0.30); no patient shared between the expert set and the main study; external check on VeriFact-BHC (Sections 3.5, 6.5, 6.7)
- Confounds and inference: length-matched E0, E1b without the citation instruction, repeated generations; equivalence of E1b's coverage tested with two one-sided tests; E2 and the clinicians' instructions reported as references, not bounds (Sections 7.4, 7.5, 7.8)
- Robustness: second threshold, second judge with conservative and inclusive rules as alternative estimates, atomic facts, overlapping windows, verifier retrieval failures, and two further generators, with conclusions stated per generator (Section 7.10)
- Question answering marked exploratory; conclusions qualified by the judge's measured validity; reproducibility appendix with model revisions, seeds and what each result needs (Appendix B)

Plan to the defense (CSUN ETD process) and requests

Milestone	Status
ETD planning form submitted; committee confirmed (Dr. Wang, chair; Dr. McIlhenny; Dr. Ebrahimi)	Done, October 9, 2026
Second draft and progress report; advisor's review received	Done, October 12 and 15, 2026
Revised draft answering the review, response memo, these slides and the description	Done, November 10, 2026
Advisor approval; Canvas upload; Turnitin similarity 20% or lower	Next
Draft to the committee with at least two weeks of review; Doodle poll with 1-hour slots	After approval
Defense on CSUN Zoom (40 min presentation, 10 min Q&A, 10 min deliberation); slides to the advisor one week before	Before the ETD deadline
Final ETD upload after revisions; committee approval in the ETD system	December 4, 2026, 5 pm

The committee approves inside the ETD system, so the defense must be finished before the December 4 upload; the remaining dates are set with the advisor. Requests to the advisor: review the revised draft and the response memo; approve the draft for distribution to the committee; advise on the defense date.